



National  
Qualifications  
2019

**X800/76/11**

**Accounting**

FRIDAY, 17 MAY

9:00 AM – 11:30 AM

**Total marks — 120**

**SECTION 1 — 80 marks**

Attempt ALL questions.

**SECTION 2 — 40 marks**

Attempt ALL questions.

You may use a calculator.

All working should be shown fully, and clearly labelled.

Write your answers clearly in the answer booklet provided. In the answer booklet you must clearly identify the question number you are attempting.

Use **blue** or **black** ink.

Before leaving the examination room you must give your answer booklet to the Invigilator; if you do not, you may lose all the marks for this paper.



Attempt ALL questions

1. Lucas plc has 3 production departments (A, B and C) and 2 service departments (D and E). The company uses a factory-wide overhead absorption rate when calculating the cost of jobs passing through the factory.

The following data relates to Job 99.

Direct material

Material X — 5,000 kg @ £10 per kg

Material Y — 2,500 kg of material Y which is provided in large batches of 10,000 kg, each 10,000 kg batch costs £80,000

Material Z — 1,000 kg @ £30 per kg.

Direct labour

A total of 24,000 labour hours will be used to complete Job 99. These will be used in departments A, B and C in the ratio of 3:2:1. Labour rates in each department are as follows

Dept A — £12 per hour

Dept B — £8 per hour

Dept C — £10 per hour.

Machine hours required to complete Job 99 are as follows

Dept A — 3,000 hours

Dept B — 12,600 hours

Dept C — 3,000 hours.

Overheads are recovered on a factory-wide basis of £3 per direct labour hour. Lucas plc applies a profit margin of 30% to all jobs.

- (a) Prepare a Job Cost Statement to calculate the selling price of Job 99.
- (b) A new Finance Director has been appointed. She has decided to use departmental absorption rates to recover overheads. These rates are to be based on the following information.

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| Overhead costs            | Total cost |
|---------------------------|------------|
| Indirect labour           | £81,800    |
| Depreciation of machinery | £72,000    |
| Heat and light            | £104,000   |
| Factory administration    | £9,000     |
| Machinery running costs   | £150,000   |
| Rent and rates            | £26,000    |

## 1. (continued)

|                             | Dept A   | Dept B   | Dept C   | Dept D  | Dept E  | Totals   |
|-----------------------------|----------|----------|----------|---------|---------|----------|
| Indirect labour             | £30,000  | £13,100  | £18,700  | £12,300 | £7,700  | £81,800  |
| Area (sq m)                 | 20,000   | 10,000   | 17,000   | 3,000   | 2,000   | 52,000   |
| No of employees             | 400      | 160      | 240      | 40      | 60      | 900      |
| Plant and machinery at cost | £180,000 | £160,000 | £240,000 | £80,000 | £60,000 | £720,000 |
| Direct machine hours        | 36,000   | 34,000   | 24,000   | 4,000   | 2,000   | 100,000  |
| Direct labour hours         | 50,000   | 24,000   | 36,000   | 6,000   | 4,000   | 120,000  |

- (i) Prepare an overhead analysis sheet to calculate the total overheads allocated and apportioned to all 5 departments. 16
- (ii) Re-apportion the total costs of Department D to the other 4 departments on the basis of direct labour hours. 3
- (iii) Re-apportion the total costs of Department E to the 3 production departments on the basis of direct machine hours. 3
- (c) (i) Calculate the overhead absorption rate applied to  
 Department A — based on direct labour hours  
 Department B — based on direct machine hours  
 Department C — based on direct labour hours. 3
- (ii) Calculate the increase or decrease in the overheads applied to Job 99 by using the new departmental overhead rates rather than the factory-wide rates. 5
- (d) Outline the impact of ‘under’ absorption of overheads on profit. 1

[Turn over]

2. Part A

Newben plc makes 3 products and in Year 10 is working at 80% capacity. The following information is available regarding the manufacture of products A, B and C.

|                         | Product A | Product B | Product C |
|-------------------------|-----------|-----------|-----------|
| Selling price per unit  | £120      | £90       | £106      |
| Sales demand in units   | 1,000     | 4,000     | 3,500     |
| Machine hours per unit  | 3         | 4         | 2         |
| Material in kg per unit | 5         | 3         | 4         |
| Labour hours per unit   | 3         | 1         | 3         |

Material cost                      £3 per kg  
 Labour                              £12 per hour  
 Variable overheads              £5 per labour hour  
 Annual fixed costs                £70,000

(a) Using the information above, for Year 10 calculate

- (i) total machine hours required
- (ii) contribution per unit
- (iii) profit for the year.

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For Year 11 Newben plc is considering the issues of both machine capacity and profit maximisation. Market research has indicated that maximum sales demand for each product will be

| Product | Sales demand (units) |
|---------|----------------------|
| A       | 4,000                |
| B       | 3,500                |
| C       | 5,000                |

In an attempt to meet this demand and maximise profits Newben plc will have to work at full capacity.

(b) Calculate for Year 11

- (i) total machine hours at full capacity
- (ii) the quantity of each product to be produced in order to maximise profit
- (iii) total contribution
- (iv) total profit if fixed costs increase by 15%.

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2. (continued)

Newben plc is considering the following proposal for Year 12.

To increase demand for Product A, Newben plc plans to reduce its selling price to £117 per unit. Demand is expected to increase by 400 units. Fixed costs will remain as Year 11 and they will continue to operate at full capacity.

- |                                                                                                                    |   |
|--------------------------------------------------------------------------------------------------------------------|---|
| (c) Calculate the expected profit for Year 12 if the changes are implemented.                                      | 6 |
| (d) Recommend to Newben plc whether they should implement the proposed change for Year 12 and justify your answer. | 1 |

2. Part B

Petroscot produces their product using 2 separate processes, refining and mixing. The mixing process combines 2 litres of Material A with 1 litre of Material B.

Cost data for the mixing process for the month of May is as follows

|                      |                                                                      |
|----------------------|----------------------------------------------------------------------|
| Materials            | 4,000 litres of Material A from the refining process at £5 per litre |
| Additional materials | ? litres of Material B at £2 per litre                               |
| Labour               | 3,000 hours of direct labour at £12 per hour.                        |

Overheads are to be charged as follows

|                    |                                          |
|--------------------|------------------------------------------|
| Variable overheads | £2.50 per litre of total input materials |
| Fixed overheads    | £6,000.                                  |

Other forecasted data

|                            |                             |
|----------------------------|-----------------------------|
| Normal loss                | 5% of total input quantity  |
| Closing work-in-progress   | 500 litres valued at £1,800 |
| Completed output to stores | 5,000 litres.               |

All losses are to be sold for £4 per litre.

- |                                                                                             |   |
|---------------------------------------------------------------------------------------------|---|
| (a) Using the information shown above, <b>prepare</b> for the month of May the              |   |
| (i) Mixing Process Account showing quantities, cost and values                              | 9 |
| (ii) Abnormal Loss Account.                                                                 | 3 |
| (b) (i) Outline <b>one</b> feature of process costing.                                      | 1 |
| (ii) In connection with process costing, distinguish between normal loss and abnormal loss. | 2 |

[Turn over

**SECTION 2 — 40 marks****Attempt ALL questions****3. Coulter and Co plc is a manufacturing business.**

Ledger balances as at 31 December Year 3 were as follows

|                                                     | <b>£000</b> |
|-----------------------------------------------------|-------------|
| Sales Revenue                                       | 1,950       |
| Factory Machinery at cost                           | 630         |
| Delivery Vehicles at cost                           | 250         |
| Inventories at 1 January Year 3:                    |             |
| Raw Materials                                       | 80          |
| Work-in-Progress                                    | 45          |
| Finished Goods                                      | 110         |
| Rates                                               | 33          |
| Production Wages                                    | 120         |
| Management Salaries                                 | 60          |
| Factory Indirect Labour                             | 111         |
| Purchase of Raw Materials                           | 200         |
| Carriage In on Raw Materials                        | 3           |
| Carriage Out                                        | 4           |
| Factory (indirect) Power                            | 10          |
| Advertising                                         | 25          |
| Repairs to Factory Machinery                        | 14          |
| Bad Debts                                           | 6           |
| Factory Heat and Light                              | 23          |
| Selling Expenses                                    | 120         |
| Royalties Paid                                      | 5           |
| Debenture Interest                                  | 10          |
| Warehouse Expenses                                  | 47          |
| Provisions for Depreciation as at 1 January Year 3: |             |
| Factory Machinery                                   | 200         |
| Delivery Vehicles                                   | 50          |
| Factory Cleaning                                    | 30          |

**3. (continued)**

The following additional information was made available at 31 December Year 3.

1. Inventories at 31 December Year 3 were as follows  
 Raw Materials — £60,000  
 Work-in-Progress — £15,000  
 Finished Goods — £90,000.
2. On 31 December Year 3, Rates of £7,000 were payable.
3. The following expenses are to be apportioned as follows

|                     | Factory | Warehouse | Admin |
|---------------------|---------|-----------|-------|
| Rates               | 80%     | —         | 20%   |
| Management Salaries | 75%     | 10%       | 15%   |

4. All Non-Current Assets are to be depreciated at the rate of 20% using the reducing balance method.
  5. Factory cleaning costs have been paid for the 15 months ending 31 March year 4.
  6. The work completed in Year 3 has been estimated to have a Market Value of £1,000,000.
- (a) (i) Prepare the Manufacturing Account for the year ended 31 December Year 3. **13**
- (ii) Prepare the Income Statement (up to Gross Profit) for the year ended 31 December Year 3. **6**
- (b) Describe the term Work-in-Progress. **1**

[Turn over

4. The Income Statement of Cromer plc for the year ended 31 December Year 4 has been completed to the Profit for the Year after Tax. The following information is also available.

|                                                    | £000         | £000         |
|----------------------------------------------------|--------------|--------------|
| Profit for the Year after Tax                      |              | 90           |
| Corporation Tax Payable                            |              | 30           |
| Unappropriated Profit 31 December Year 3           |              | 36           |
| 400,000 6% Preference Shares £1 each               |              | 400          |
| 400,000 Ordinary Shares £1 each                    |              | 400          |
| VAT                                                | 28           |              |
| Provision for Doubtful Debts 31 December Year 4    |              | 6            |
| Trade Receivables                                  | 44           |              |
| Trade Payables                                     |              | 32           |
| Share Premium                                      |              | 150          |
| Cash and Cash Equivalents                          | 7            |              |
| Goodwill                                           | 20           |              |
| Preliminary Expenses                               | 60           |              |
| Property (at cost)                                 | 690          |              |
| Office Equipment (at cost)                         | 300          |              |
| Motor Vehicles (at cost)                           | 180          |              |
| Provisions for Depreciation at 31 December Year 4: |              |              |
| Office Equipment                                   |              | 172          |
| Motor Vehicles                                     |              | 54           |
| Closing Inventory                                  | 26           |              |
| Admin Expenses Payable                             |              | 4            |
| Selling Expenses Receivable                        | 4            |              |
| Ordinary Dividend Paid                             | 15           |              |
|                                                    | <b>1,374</b> | <b>1,374</b> |

#### Notes at 31 December Year 4

- At 31 December Year 4 a bonus issue of 50,000 Ordinary Shares was made. This issue was financed by a transfer from the Share Premium Account. This has yet to be recorded in the accounts.
- Property is revalued at £720,000.
- Write-off the Preliminary Expenses by transfer from the Share Premium Account.
- Goodwill is to be written down by £8,000.
- The preference dividend has been paid by cheque, but no transactions have been made.



**4. (continued)**

- (a) The Income Statement has been completed to determine Profit for the Year after Tax.

Prepare

- |                                                                                                                        |           |
|------------------------------------------------------------------------------------------------------------------------|-----------|
| (i) the remainder of the Income Statement to determine the Unappropriated Profit for the year ended 31 December Year 4 | <b>3</b>  |
| (ii) a Statement of Financial Position as at 31 December Year 4.                                                       | <b>15</b> |
| (b) Compare Debentures with Preference Shares.                                                                         | <b>2</b>  |

**[END OF QUESTION PAPER]**

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